Week 1 complete solutions

Problem-sheet solutions

Each solution states the intended method, relevant conditions, mark allocation and common mistakes. Award method credit after a purely arithmetic slip when the setup remains valid.

Question 1 — sets and functions

`W01-Q01-PA (M1)` **Method and conditions.** List the two subsets of the stated universal set Ω, then apply the definitions of intersection, union, complement and subset.

We have

$$A = \{2, 4, 6, 8, 10, 12\},$$
$$B = \{1, 2, 3, 4, 6, 12\}.$$

It follows that

$$A \cap B = \{2, 4, 6, 12\},$$
$$A \cup B = \{1, 2, 3, 4, 6, 8, 10, 12\},$$
$$A^c = \{1, 3, 5, 7, 9, 11\}.$$

The statement $B \subseteq A$ is false. For example, $1 \in B$ but $1 \notin A$.

Marks (5). Intersection 1; union 1; complement 1; correct subset decision 1; valid counterexample 1.

Common mistakes. Confusing union with intersection; taking the complement relative to a set other than Ω; giving a conclusion about the subset relation without a counterexample.

`W01-Q01-PB (M1)` **Method and conditions.** Test injectivity and surjectivity against the stated domain/codomain, then distinguish implication from equivalence and use an odd-square contradiction for the reverse implication.

Every input having exactly one output makes f a function. Injectivity instead requires distinct inputs to have distinct outputs. Here,

$$f(-2) = f(2) = 4 \quad \text{and} \quad f(-1) = f(1) = 1,$$

so f is not injective. Its range is $\{0, 1, 4\}$, which equals the stated codomain. Therefore f is surjective.

The parity relation is $P \Leftrightarrow Q$. If $n = 2k$, then $n^2 = 4k^2$, so $P \Rightarrow Q$. Conversely, suppose n^2 is even but n is odd. Then $n = 2k + 1$, so

$$n^2 = 4k^2 + 4k + 1,$$

which is odd, a contradiction. Thus $Q \Rightarrow P$.

Marks (4). Valid collision and non-injectivity conclusion, 1; range and surjectivity conclusion, 1; correct equivalence relation, 1; valid contradiction for $Q \Rightarrow P$, 1.

Common mistakes. Treating "one output per input" as injectivity; checking surjectivity without the stated codomain; giving only $P \Rightarrow Q$; assuming the reverse implication without justification.

Question 2 — quantifiers and induction

`W01-Q02-PA (M1)` **Method and conditions.** Negate both the universal quantifier and the strict inequality, then use a property of real squares.

The negation is

$$\exists x \in \mathbb{R} \quad \text{such that} \quad x^2 + 1 \leq 0.$$

The original statement is true. For every real x, $x^2 \geq 0$, so $x^2 + 1 \geq 1 > 0$.

Marks (5). Existential quantifier 1; correctly negated inequality 1; truth value 1; use of $x^2 \geq 0$ 1; complete conclusion for every real x 1.

Common mistakes. Retaining the universal quantifier; writing < 0 instead of ≤ 0; citing a few numerical cases as a universal proof.

`W01-Q02-PB (M1, M2)` **Method and conditions.** Prove the identity for all integers $n \geq 1$ by establishing a base case and the implication from n to $n + 1$.

For $n = 1$, the left-hand side is $3(1) - 1 = 2$, while the right-hand side is $1(3 + 1)/2 = 2$. The base case holds.

Assume for some $n \geq 1$ that

$$\sum_{k=1}^{n}(3k - 1) = \frac{n(3n + 1)}{2}.$$

Then

$$\begin{aligned}
\sum_{k=1}^{n+1}(3k - 1) &= \sum_{k=1}^{n}(3k - 1) + 3(n + 1) - 1 \\
&= \frac{n(3n + 1)}{2} + 3n + 2 \\
&= \frac{3n^2 + 7n + 4}{2} \\
&= \frac{(n + 1)(3n + 4)}{2} \\
&= \frac{(n + 1)(3(n + 1) + 1)}{2}.
\end{aligned}$$

This is the required formula at $n + 1$. Therefore the identity holds for all $n \in \mathbb{N}$ with $n \geq 1$.

Marks (8). Base-case calculation 2; induction hypothesis 1; correct next summand 1; substitution of the hypothesis 1; algebra and factorisation 2; induction conclusion 1.

Common mistakes. Omitting the base case; assuming the result at $n+1$; adding $3n-1$ rather than $3(n+1)-1$; reaching the target expression without stating the induction conclusion.

Question 3 — logarithms and a declining benefit stream

`W01-Q03-PA (M2)` **Method and conditions.** State the positivity conditions for the logarithm arguments, combine the logarithms, and retain only admissible roots.

The domain requires $x - 1 > 0$ and $x + 2 > 0$, so $x > 1$. On this domain,

$$\log((x - 1)(x + 2)) = \log 18.$$

Because log is injective on its positive domain,

$$(x-1)(x+2) = 18,$$

and hence

$$x^2 + x - 20 = 0 = (x-4)(x+5).$$

The candidates are $x = 4$ and $x = -5$. Only $x = 4$ satisfies $x > 1$, so the solution is $x = 4$.

Marks (6). Domain $x > 1$ 1; logarithm product rule 1; equation after injectivity 1; quadratic solution 2; rejection of -5 and final answer 1.

Common mistakes. Manipulating logarithms before enforcing positivity; writing a sum rather than a product inside the logarithm; keeping the inadmissible root.

`W01-Q03-PB (M2)` **Method and conditions.** Model the stream as a geometric series. An infinite geometric series converges when the absolute value of its common ratio is less than one.

The index range in the product is $i = 1, 2, 3, 4$, so

$$\prod_{i=1}^{4} r^i = r^1 r^2 r^3 r^4 = r^{1+2+3+4} = r^{10}.$$

The benefit at t is $100(3/4)^t$. Because $|3/4| < 1$, the infinite total is

$$\sum_{t=0}^{\infty} 100 \left(\frac{3}{4}\right)^t = \frac{100}{1 - 3/4} = 400.$$

The cumulative benefit through year N is

$$S_N = 100 \frac{1 - (3/4)^{N+1}}{1 - 3/4} = 400 \left(1 - \left(\frac{3}{4}\right)^{N+1}\right).$$

The threshold $S_N \geq 350$ is equivalent to

$$\left(\frac{3}{4}\right)^{N+1} \leq \frac{1}{8}.$$

Taking logarithms gives

$$(N+1)\log(3/4) \leq \log(1/8).$$

Since $\log(3/4) < 0$, division reverses the inequality:

$$N + 1 \geq \frac{\log(1/8)}{\log(3/4)} \approx 7.23.$$

Thus the smallest possible integer is $N + 1 = 8$, so $N = 7$. The total through $t = 0, \ldots, 7$ is the first one to reach 350 units.

Marks (8). Indexed-product range/expansion, 1; product simplification, 1; geometric model with convergence and infinite total, 2; finite-sum expression and threshold inequality, 2; logarithmic solution with reversed inequality and smallest integer, 2.

Common mistakes. Starting the product at $i = 0$; multiplying rather than adding exponents; starting the series at $t = 1$; omitting $|r| < 1$; forgetting that division by $\log(3/4) < 0$ reverses the inequality.

Question 4 — a logarithmic function

`W01-Q04-PA (M2)` **Method and conditions.** First impose strict positivity of the logarithm argument. Then use that log is strictly increasing.

The domain condition is

$$9 - x^2 > 0,$$

so the domain is $(-3, 3)$. Within this domain,

$$\begin{aligned} \log(9 - x^2) &\geq \log 5 \\ 9 - x^2 &\geq 5 \\ x^2 &\leq 4. \end{aligned}$$

Thus $-2 \leq x \leq 2$. This interval is contained in the domain, so the required set is $[-2, 2]$.

Marks (6). Positivity condition 1; domain $(-3, 3)$ 1; justified use of monotonicity 1; reduction to $x^2 \leq 4$ 1; solution of that inequality 1; domain intersection and interval answer 1.

Common mistakes. Including $x = \pm 3$; reversing the inequality even though log is increasing; solving the algebraic inequality without checking the logarithm's domain.

Question 5 — derivatives, an integral and marginal cost

`W01-Q05-PA (M3)` **Method and conditions.** Use the product rule on the domain $x > 0$, then differentiate once more. Evaluate the definite integral from an antiderivative obtained by integration by parts.

The derivatives are

$$\begin{aligned} f'(x) &= 2x \log x + x, \\ f''(x) &= 2 \log x + 3. \end{aligned}$$

Integration by parts gives

$$\int \log x \, \mathrm{d}x = x \log x - x + C.$$

Therefore

$$\int_1^e \log x \, \mathrm{d}x = [x \log x - x]_1^e = 0 - (-1) = 1.$$

Marks (6). Product-rule structure for f' 1; correct f' 1; correct differentiation to f'' 2; antiderivative 1; endpoint evaluation 1.

Common mistakes. Omitting a product-rule term; using $(\log x)' = \log x / x$; evaluating the lower endpoint as zero.

`W01-Q05-PB (M3)` **Method and conditions.** Interpret the derivative as a local rate under the given differentiable approximation, and compare it with the finite change over a whole unit.

We have $C'(q) = 4 + q$, so $C'(6) = 10$ euros per unit. This is the instantaneous marginal cost at output 6: for a small increase Δq, the cost increase is approximately $10\Delta q$ euros.

The exact increase from 6 to 7 is

$$C(7) - C(6) = 4(7 - 6) + \frac{1}{2}(7^2 - 6^2) = 4 + \frac{13}{2} = 10.5$$

euros. The colleague's claim is therefore not exact; 10 euros is a local linear approximation, while the model gives 10.5 euros for the full one-unit increase.

Marks (4). Derivative calculation 1; local marginal interpretation with units 1; exact finite difference 1; correct assessment of the claim 1.

Common mistakes. Treating a derivative as an exact discrete increment; omitting euros per unit from the derivative; computing $C(7) - C(6)$ as $C'(7)$.

Question 6 — Taylor approximation

`W01-Q06-PA (M3)` **Method and conditions.** Centre both approximations at $a = 1$, which is in the domain $x > 0$. Use the function and derivative values at the centre.

For $f(x) = \log x$,

$$f(1) = 0, \qquad f'(x) = \frac{1}{x}, \qquad f'(1) = 1, \qquad f''(x) = -\frac{1}{x^2}, \qquad f''(1) = -1.$$

The first-order approximation is

$$T_1(x) = x - 1,$$

and the second-order approximation is

$$T_2(x) = (x - 1) - \frac{1}{2}(x - 1)^2.$$

At $x = 1.2$, these give

$$\log 1.2 \approx T_1(1.2) = 0.2$$

and

$$\log 1.2 \approx T_2(1.2) = 0.2 - \frac{1}{2}(0.2)^2 = 0.18.$$

Marks (8). Function and derivative values at the centre 2; first-order formula 2; second-order formula including the factor 1/2 2; two numerical approximations 2.

Common mistakes. Centring at zero; omitting the factorial factor; using a positive quadratic term; substituting x instead of $x - 1$.

`W01-Q06-PB (M3)` **Method and conditions.** Compare absolute, rather than signed, errors and interpret only what this pointwise calculation establishes.

Using the supplied comparison value, the first-order absolute error is

$$|0.2 - 0.1823| = 0.0177,$$

whereas the second-order absolute error is

$$|0.18 - 0.1823| = 0.0023.$$

The second-order approximation is substantially closer at $x = 1.2$. This single comparison illustrates the benefit of the quadratic term near the centre $a = 1$; it does not prove that a higher-order approximation is better at every point or on every domain.

Marks (3). First absolute error 1; second absolute error 1; appropriately limited interpretation 1.

Common mistakes. Comparing signed errors; claiming that the second-order approximation is exact; turning one local comparison into a global theorem.

Question 7 — optimisation on an interval

`W01-Q07-PA (M3)` **Method and conditions.** A continuous function on the closed interval $[0, 12]$ attains a global maximum. Compare all feasible stationary points with both endpoints.

The derivatives are

$$\pi'(q) = 18 - 2q \quad \text{and} \quad \pi''(q) = -2.$$

The stationary point is $q = 9$, which is feasible. Since $\pi''(q) = -2 < 0$, the profit function is strictly concave. The candidate values are

$$\pi(0) = -32, \qquad \pi(9) = 162 - 81 - 32 = 49, \qquad \pi(12) = 216 - 144 - 32 = 40.$$

The global profit-maximising output is $q = 9$, and the maximum profit is 49.

Marks (8). Derivative and stationary point 2; feasibility and second-derivative information 1; three candidate values 3; comparison 1; output and maximum profit 1.

Common mistakes. Omitting the endpoints; treating the first-order condition as sufficient without curvature or comparison; reporting only the maximising output.

`W01-Q07-PB (M3)` **Method and conditions.** Identify the gap in a first-order-condition-only argument and then use the evidence specific to this function and feasible set.

On a closed interval, an interior stationary point must be compared with the endpoints; in general it might be a minimum or fail to be globally best. Here the endpoint comparison shows that $\pi(9)$ exceeds both endpoint values. Moreover, strict concavity means there is at most one maximiser. These facts validate the global conclusion.

Marks (2). Missing endpoint or sufficiency check identified 1; valid problem-specific global justification 1.

Common mistakes. Saying every stationary point is a maximum; mentioning endpoints without comparing their values; invoking concavity without noting its sign.

Question 8 — probability as set algebra

`W01-Q08-PA (S1)` **Method and conditions.** Apply inclusion–exclusion, then use complements and the disjoint pieces of the union.

First,

$$\mathbb{P}(A \cup B) = 0.58 + 0.47 - 0.31 = 0.74.$$

Therefore

$$\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A \cup B) = 0.26.$$

The event that exactly one occurs is $(A \quad B) \cup (B \quad A)$, a disjoint union. Its probability is

$$\mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = 0.74 - 0.31 = 0.43.$$

Marks (8). Inclusion–exclusion setup and union value 2; complement setup and neither value 2; correct event or formula for exactly one 2; exactly-one value 2.

Common mistakes. Adding the two marginal probabilities without removing overlap; treating "neither" as the complement of the intersection; retaining the intersection in the exactly-one event.

`W01-Q08-PB (S1)` **Method and conditions.** Interpret addition as counting event regions.

The sum $0.58 + 0.47$ counts every outcome in $A \cap B$ once as part of A and again as part of B. The intersection has probability 0.31, so it must be subtracted once. Hence

$$\mathbb{P}(A \cup B) = 0.58 + 0.47 - 0.31 = 0.74.$$

Marks (2). Double-counted intersection identified 1; correct event-level correction and result 1.

Common mistakes. Saying only that probabilities cannot exceed one; subtracting the intersection twice; assuming the events are disjoint despite the supplied overlap.

Question 9 — a panel with a chair

`W01-Q09-PA (S1)` **Method and conditions.** Count unrestricted unordered panels with a designated chair, then subtract the outcomes containing both prohibited applicants.

Without the restriction, there are

$$\binom{8}{3} \times 3 = 56 \times 3 = 168$$

panel-and-chair outcomes. If Ada and Bruno both sit, the third member can be any of the other six applicants, and any of the three panel members can chair. Thus there are

$$6 \times 3 = 18$$

forbidden outcomes. The required number is

$$168 - 18 = 150.$$

Marks (7). Unrestricted panel count 1; chair designation 1; forbidden third-member count 1; forbidden chair count 1; correct subtraction setup 2; final total 1.

Common mistakes. Ordering all three panel members; excluding every panel containing Ada or Bruno; forgetting that each forbidden panel has three possible chairs.

Question 10 — representation across proposal fields

`W01-Q10-PA (S1)` **Method and conditions.** All $\binom{12}{4}$ unordered selections are equally likely. A selection representing three fields must have field counts $2, 1, 1$.

The total number of selections is

$$\binom{12}{4} = 495.$$

Counting by the field that supplies two proposals gives

$$\begin{aligned}
\text{two theory:} &\quad \binom{5}{2}\binom{4}{1}\binom{3}{1} = 120,\\
\text{two applied:} &\quad \binom{5}{1}\binom{4}{2}\binom{3}{1} = 90,\\
\text{two history:} &\quad \binom{5}{1}\binom{4}{1}\binom{3}{2} = 60.
\end{aligned}$$

Thus there are $120 + 90 + 60 = 270$ favourable selections, and

$$\mathbb{P}(\text{all three fields represented}) = \frac{270}{495} = \frac{6}{11}.$$

For a check, let E_T, E_A and E_H be the events that theory, applied economics and economic history, respectively, are absent. The counts are

$$|E_T| = \binom{7}{4} = 35, \qquad |E_A| = \binom{8}{4} = 70, \qquad |E_H| = \binom{9}{4} = 126.$$

The pairwise-intersection counts are

$$|E_T \cap E_A| = \binom{3}{4} = 0, \quad |E_T \cap E_H| = \binom{4}{4} = 1, \quad |E_A \cap E_H| = \binom{5}{4} = 5,$$

and the triple intersection is empty. Inclusion–exclusion therefore gives

$$|E_T \cup E_A \cup E_H| = 35 + 70 + 126 - 0 - 1 - 5 = 225.$$

The complement has $495 - 225 = 270$ selections, agreeing with the direct count and hence with the probability $6/11$.

Marks (10). Total sample-space count 1; $2, 1, 1$ case structure 1; three direct case counts 3; favourable total and probability 2; correct complement inclusion–exclusion count 2; reconciliation 1.

Common mistakes. Treating the draw order as relevant; omitting one of the three doubled-field cases; forgetting pairwise intersections in the complement count; using a denominator other than $\binom{12}{4}$.

Self-test solutions and error check

Solutions and error check

Solution to `W01-ST01-PA (M1)` Method. Distinguish the definition of a function from the definitions of injectivity and surjectivity, using the stated domain and codomain.

Every input has exactly one output, so the rule defines a function. It is not injective because

$$f(-1) = f(1) = 1$$

for two distinct inputs. Its set of output values is

$$\{f(-1), f(0), f(1), f(2)\} = \{0, 1, 4\},$$

which equals the stated codomain. Therefore f is surjective.

Marks (5). Function/injectivity distinction, 1; valid collision, 1; correct non-injectivity conclusion, 1; output-set check, 1; correct surjectivity conclusion, 1.

Common mistakes. Calling every function injective; checking only adjacent inputs; ignoring the stated codomain when testing surjectivity.

Solution to `W01-ST02-PA (M2)` Method. Enforce positivity of both logarithm arguments before using the product rule.

The real-log domain requires

$$x - 2 > 0 \quad \text{and} \quad x + 1 > 0,$$

so $x > 2$. On this domain,

$$\log((x-2)(x+1)) = \log 10.$$

Because log is one-to-one,

$$(x-2)(x+1) = 10,$$

and hence

$$x^2 - x - 12 = (x-4)(x+3) = 0.$$

The algebraic candidates are 4 and -3. Only 4 satisfies $x > 2$, so the solution is $x = 4$.

Marks (5). Complete domain, 1; valid logarithm combination, 1; correct quadratic, 1; both algebraic candidates, 1; domain check and final answer, 1.

Common mistakes. Combining logarithms before stating the domain; adding the two arguments; retaining the inadmissible negative root.

Solution to `W01-ST03-PA (M3)` Method. Find every feasible stationary point and compare it with both endpoints.

$$\pi'(q) = 8 - 2q, \qquad \pi''(q) = -2.$$

The stationary point is $q = 4$, and the negative second derivative shows strict concavity. The candidate values are

$$\pi(0) = 0, \qquad \pi(4) = 16, \qquad \pi(6) = 12.$$

Therefore the global maximiser is $q = 4$, and the maximum value is 16.

Marks (5). Correct derivative and stationary point, 2; curvature check, 1; both endpoint values, 1; global maximiser and value, 1.

Common mistakes. Stopping at the first-order condition; omitting one endpoint; reporting the maximiser without the maximum value.

Solution to `W01-ST04-PA (S1)` Method. Use inclusion–exclusion, then take the relevant complement and remove the overlap when counting exactly one event.

$$\mathbb{P}(A \cup B) = 0.6 + 0.5 - 0.3 = 0.8.$$

Therefore,

$$\mathbb{P}(\text{neither}) = 1 - \mathbb{P}(A \cup B) = 0.2.$$

Exactly one event occurs on the union excluding the intersection, so

$$\mathbb{P}(\text{exactly one}) = \mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = 0.8 - 0.3 = 0.5.$$

Marks (5). Correct union setup and value, 2; correct complement, 1; correct exactly-one setup and value, 2.

Common mistakes. Adding probabilities without subtracting the overlap; taking the complement of the intersection for neither; retaining the intersection in the exactly-one event.

Record the first cause of each error under `M1`, `M2`, `M3` or `S1`, then reattempt that part from a blank page in two or three days.